Red Hat OpenShift AI Self-Managed 3.0

Enabling the model registry component

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Abstract

Use the model registry to register, version, and manage AI model versions, and use the model catalog to discover available generative AI models for deployment.

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PREFACE

As an OpenShift cluster administrator, before data scientists and OpenShift AI administrators in your organization can work with the model registry and model catalog, you must ensure that the model registry component is enabled in OpenShift AI.

NOTE

The model registry component is enabled by default in a new OpenShift AI 3.0 installation. However, if the model registry was not enabled in a previous version of OpenShift AI, you must enable this component after upgrading to OpenShift AI 3.0.

CHAPTER 1. OVERVIEW OF THE MODEL CATALOG AND MODEL REGISTRIES

The model catalog provides a curated library where data scientists and AI engineers can discover and evaluate the available generative AI (gen AI) models to find the best fit for their use cases.

A model registry acts as a central repository for administrators, data scientists, and AI engineers to register, version, and manage the lifecycle of AI models before configuring them for deployment. A model registry is a key component for AI model governance.

1.1. MODEL CATALOG

Data scientists and AI engineers can use the model catalog to discover and evaluate the gen AI models that are available and ready for their organization to register, deploy, and customize.

The model catalog provides models from different providers that you can search, discover, and evaluate before you register models in a model registry and deploy them to a model serving runtime. Third-party gen AI models are benchmarked by Red Hat for performance and quality by using open-source evaluation datasets. You can compare performance metrics for specific hardware configurations and determine the most suitable option for deployment.

OpenShift AI provides a default model catalog, which includes models from providers such as Red Hat, IBM, Meta, Nvidia, Mistral AI, and Google. OpenShift AI administrators can configure the available repository sources for models displayed in the model catalog.

For more information about how data scientists and AI engineers can use the model catalog, see Working with the model catalog.

1.2. MODEL REGISTRY

A model registry is an important component in the lifecycle of an artificial intelligence/machine learning (AI/ML) model, and is a vital part of any machine learning operations (MLOps) platform or workflow. A model registry acts as a central repository, storing metadata related to machine learning models from development to deployment. This metadata ranges from high-level information like the deployment environment and project, to specific details like training hyperparameters, performance metrics, and deployment events.

A model registry acts as a bridge between model experimentation and serving, offering a secure, collaborative metadata store interface for stakeholders in the ML lifecycle. Model registries provide a structured and organized way to store, share, version, deploy, and track models.

OpenShift AI administrators can create model registries in OpenShift AI and grant model registry access to data scientists and AI engineers. For more information, see Managing model registries.

Data scientists and AI engineers with access to a model registry can use it to store, share, version, deploy, and track models. For more information, see Working with model registries.

CHAPTER 2. ENABLING THE MODEL REGISTRY COMPONENT

Before data scientists and AI engineers in your organization can work with the model registry and model catalog, you must ensure that the **modelregistry** component is enabled in OpenShift AI.

> **NOTE**
>
> The **modelregistry** component is enabled by default in a new OpenShift AI 3.0 installation. However, if the model registry was not enabled in a previous version of OpenShift AI, you must enable this component after upgrading to OpenShift AI 3.0.

Prerequisites

- You have cluster administrator privileges for your OpenShift cluster.
- You have access to the data science cluster.
- You have installed the Red Hat OpenShift AI Operator on your OpenShift cluster.
- You have sufficient resources. For more information about the minimum resources required to use OpenShift AI, see Installing and deploying OpenShift AI (for disconnected environments, see Deploying OpenShift AI in a disconnected environment).

Procedure

1. In the OpenShift console, click **Operators → Installed Operators**.
2. Search for the **Red Hat OpenShift AI Operator** version 2.14+, and then click the Operator name to open the Operator details page.
3. Click the **Data Science Cluster** tab.
4. Click the default instance name (for example, **default-dsc**) to open the instance details page.
5. Click the **YAML** tab to show the instance specifications.
6. Find the **spec.components** section, and then add or update it to include the following **modelregistry** component entry, with the **managementState** field set to **Managed**, and the **registriesNamespace** field set to **rhoai-model-registries**:

   ```
   modelregistry:
     managementState: Managed
     registriesNamespace: rhoai-model-registries
   ```

7. Click **Save**.

Verification

- Confirm that the model registry namespace was created successfully:
  a. In the OpenShift console, click **Home → Projects**.
  b. Confirm that the **rhoai-model-registries** namespace is displayed in the **Projects** drop-down list.

- Check the status of the **model-registry-operator-controller-manager** pod:
  a. In the OpenShift console, from the **Project** list, select **redhat-ods-applications**.
  b. Click **Workloads** → **Deployments**.
  c. Search for the **model-registry-operator-controller-manager** deployment.
  d. Check the status:
    i. Click the deployment name to open the deployment details page.
    ii. Click the **Pods** tab.
    iii. View the pod status.
    When the status of the **model-registry-operator-controller-manager-*<pod-id>*** pod is **Running**, the pod is ready to use.

Next step

OpenShift AI administrators can create, delete, and manage permissions for model registries. For more information, see Managing model registries.